

Julian Martinez

📍 Rennes, France ✉ julianfm7@gmail.com ☎ +33 7 87 31 46 86 🌐 julianmartinezf7 🔄 PWP

Extract

Data scientist, with a solid background in mathematics and skilled in Probability/Statistics and Mathematical Modelling. I have transitioned and worked on different paths within Machine Learning. Some principles I like: *Techniques should serve the problem and not the other way around; deep understanding before deep learning.* I enjoy to use my skill-set at the service of industrial problems.

Current Position

Consultant Data Scientist, Freelance

Since Jun 2023

I am currently in a parental leave. As a side project, I have been pursuing structured learning of GenAI and GraphRag within Neo4j.

- **Dynamic Pricing:** Within a project of ESCP Business School, in partnership with [AO Hostels](#). Causal Analysis of Pricing Policy Impact (DiD – Staggered Adoption) Prepared panel data and applied difference-in-differences modeling to estimate the causal impact of a new pricing policy in the accommodation industry. Focused on data structuring, model specification, validation of assumptions, and interpretation of policy effects. Elasticity Inference from Observational Data. Reversed engineering of the pricing policy by using ML techniques.

Methodology: CI, ML, EDA, TS | Tech Stack: MLflow, Polars, Scikit-learn, CausalML, EconML, R, Kedro

- **Structured Bibliographic Data Extraction:** Built an end-to-end pipeline to extract structured bibliographic data from unstructured sources like Wikipedia and web pages. Used large language models for summarization and information extraction, enabling automated population of database fields from free-form text.

Methodology: LLM, NLP, EDA | Tech Stack: OpenAI GPT, SpaCy, Pandas, Gazpacho

Former Positions

Curriculum Engineer, [DeepLearning.AI](#) – Remote

Apr 2025 - Nov 2025

- Designed project-based learning experiences in PyTorch and modern ML workflows for large-scale educational programs.
- Collaborate with Curriculum Developers and Product Managers to ensure alignment between educational content and industry needs.
- Develop feedback-driven autograders and assessment tools to evaluate learner progress and understanding.

Methodology: Deep Learning, Computer Vision, NLP | Tech Stack: PyTorch, Git, Optuna, Docker

ML Postdoc Researcher, Orange – Cesson-Sevigne, France

Dec 2021 - Mar 2023

- Topic: Assessing Trust in a digital ecosystem by using Reinforcement Learning. I have shortly participated in a project of anomaly detection for logs.

Methodology: Reinforcement Learning, Anomaly Detection | Tech Stack: Stable Baselines3, PyTorch, Git

Consultant Data Scientist, [Aristas](#) – Buenos Aires, Argentina

Jan 2020 - Dec 2021

- Mainly in two projects: *Artificial Fragrances Generation* and *Sports Ticket Sales Forecasting*.
- Designing and implementing statistical and machine learning algorithms. Developing project proposals for clients. Meetings preparation and execution. Supervision of junior datascientists.

Methodology: GANs, VAE, Clustering, Time Series, EDA, Visualization, Supervised Learning | Tech Stack: Python, SQL, PyTorch, Scikit-learn

Professor and Researcher, University of Buenos Aires / CONICET / University of San Andres – Buenos Aires, Argentina Mar 2014 - Dec 2021

I summarize my experience in academia here, which includes research and teaching activities.

* Research: I have done research in Statistical Mechanics / Probability and Geosciences (for more details, see [Google Scholar](#)).

* Invited Professor at IMPA for the course [Continuum Percolation](#).

* Teaching: I have taught different courses in the areas of Probability, Statistics, Data Science and Machine Learning for different audiences (engineering students, economists, and professionals). I have supervised undergraduate and graduate students in their research projects and theses.

Coordinated and taught a five-course Probability and Statistics program, leading a team of 15 instructors and teaching assistants. Responsible for curriculum planning, examinations, student support, and overall academic coordination.

I created and taught two advanced courses for the Phd program in Engineering at the University of Buenos Aires: Introduction to Statistical Learning and [Reinforcement Learning](#).

Education

Leiden University, Ph.D. in Mathematics – Netherlands Sep 2010, Mar 2014
Dynamical Gibbs Non-Gibbs transitions and Brownian Percolation. (Under the supervision of [Prof.Dr. Frank den Hollander](#) and [Prof.Dr. Roberto Fernandez](#)).

I have been granted with an *Erasmus Mundus/BAPE scholarship*. In 2014, I have received the [Francisco Aranda Ordaz Award](#) to the best thesis in Probability written by a student from Latin America (Sponsored by the Bernoulli Society).

University of Buenos Aires, M.Sc. in Mathematics – Argentina Mar 2008, Mar 2010
Hydrodynamical limit for Simple Symmetric Exclusion Process (under the supervision of [Prof. Dr. Mariela Sued](#)).

Recent Technical Development

Causal Machine Learning Martin Huber
Econometric Society short course

Unstructured Data for Economics Stephen Hansen
Econometric Society short course

GraphRag Neo4j
Structured learning of GenAI and GraphRag within Neo4j

Languages

Spanish (native) | English (prof.) | French (fluent) | Portuguese (fluent)

References

Matthieu Jonckheere matthieu.jonckheere@gmail.com
LAAS (CNRS, Toulouse) (Director of Research) / [mAledge](#) (Co-Founder) / [Aristas](#) (Co-Founder)

Simon Bécot simon.becot@orange.com
Orange Business Services (Directeur de programme)

Yamila Barrera
[Mercado Libre](#) (Data Science Technical Leader)